

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

GRU Multi-Step Training Report

Report Generated: 2025-04-13 23:00:59

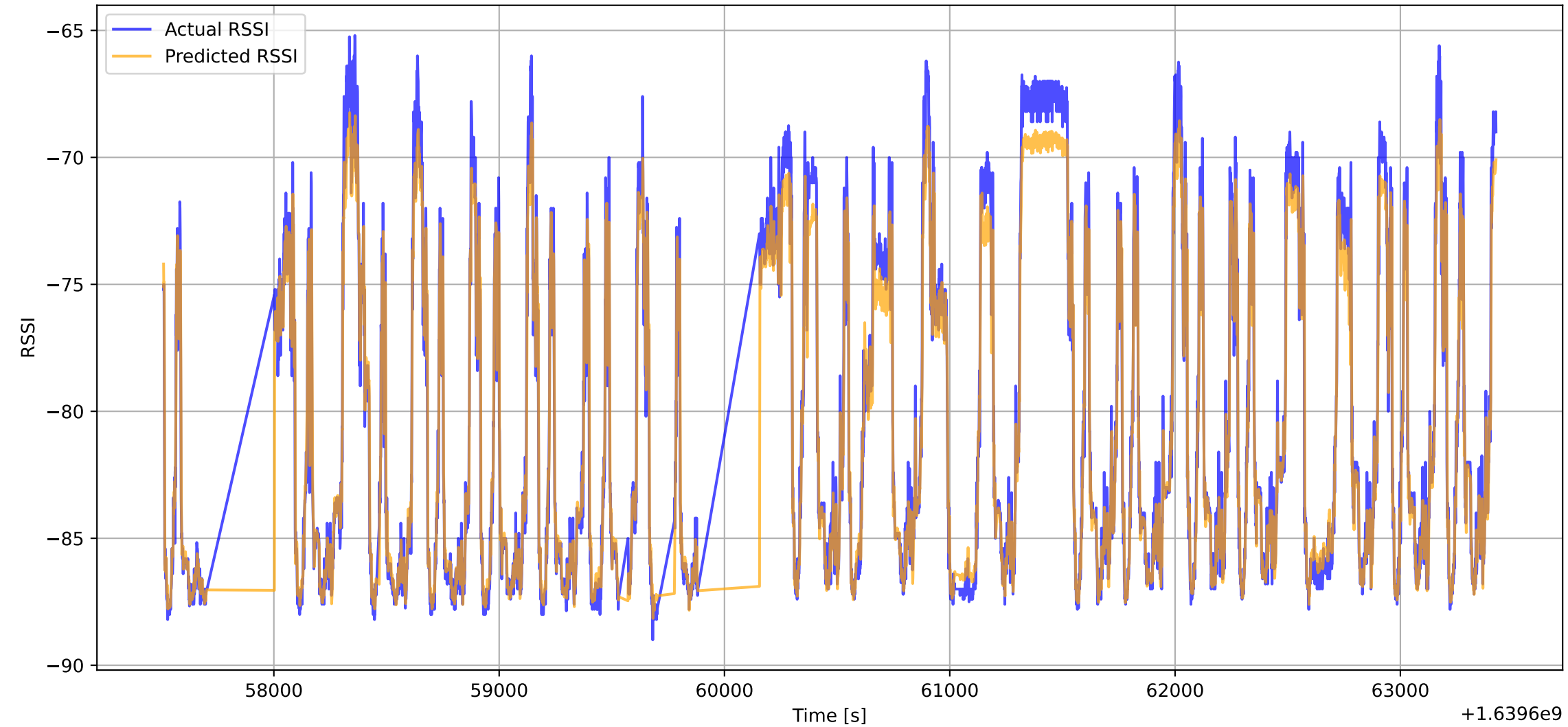
Best Hyperparameters:

Number of Layers: 2
Activation: tanh
Optimizer: Nadam
Dropout Rate: 0.20

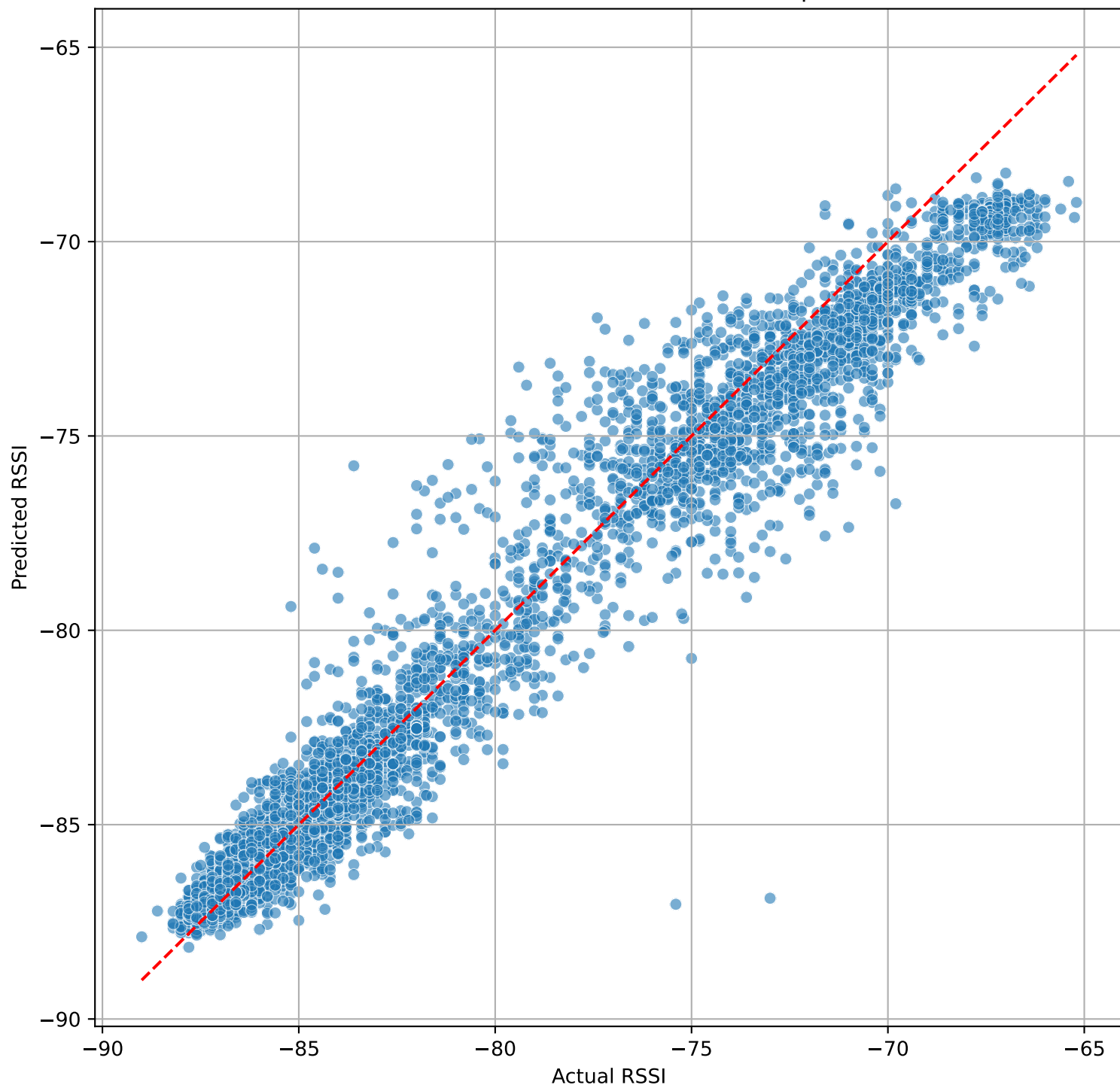
Performance Metrics (Multi-Step Prediction):

MAE: 0.9711
MSE: 1.9305
RMSE: 1.3894
R2 Score: 0.9562

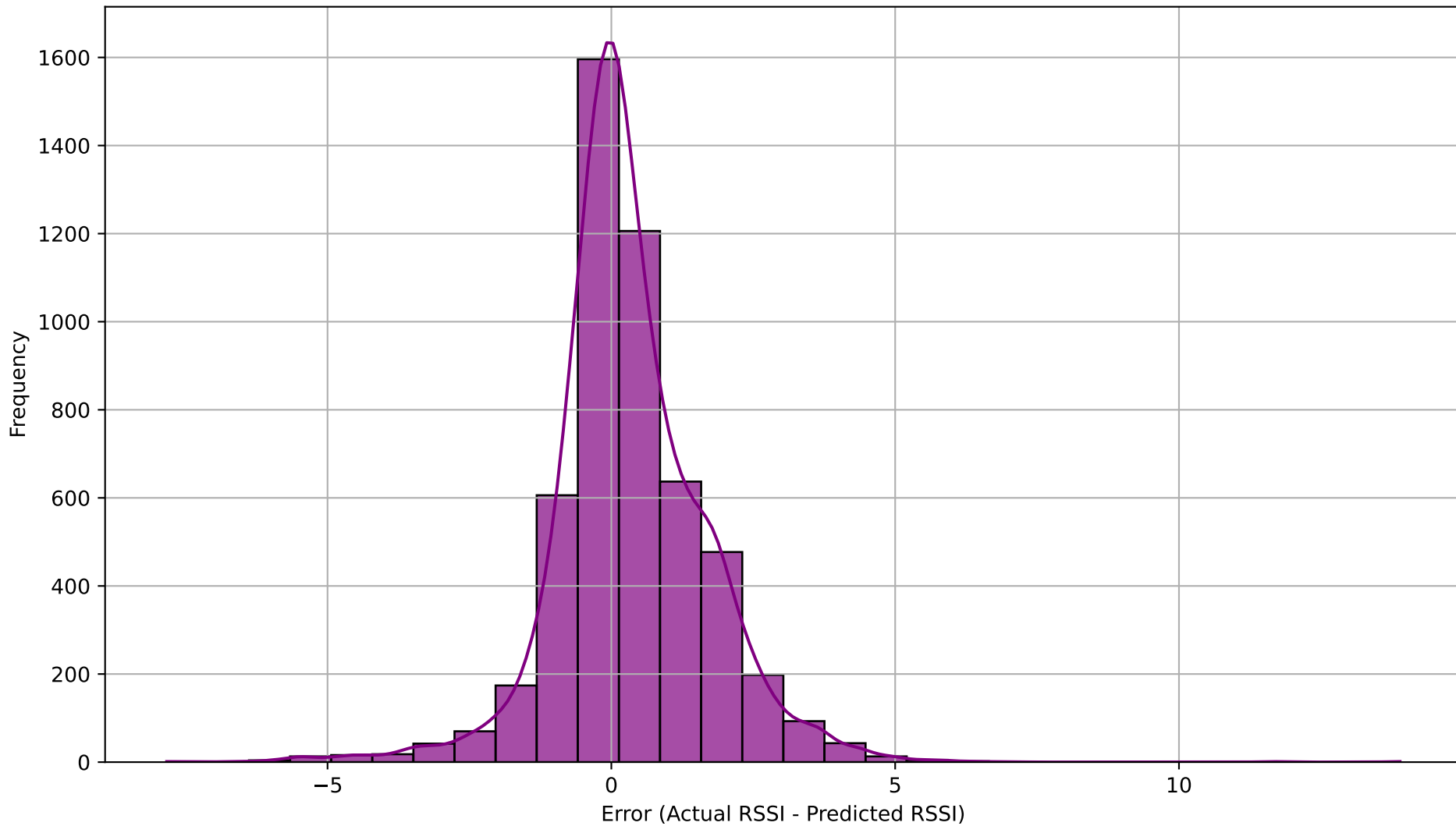
Actual vs Predicted RSSI (First Step)



Actual vs Predicted RSSI (First Step)



Distribution of Prediction Errors (First Step)



Predicted RSSI in Spatial Context (First Step)

